

On Calibration of Modern Neural Networks

Guo, Pleiss, Sun & Weinberger (2017), ICML · tags: confidence, entropy, calibration, gate

VERBATIM EXCERPT (source: arxiv.org/abs/1706.04599)

"Confidence calibration – the problem of predicting probability estimates representative of the true correctness likelihood – is important for classification models in many applications. We discover that modern neural networks, unlike those from a decade ago, are poorly calibrated."

"Through extensive experiments, we observe that depth, width, weight decay, and Batch Normalization are important factors influencing calibration."

"On most datasets, temperature scaling – a single-parameter variant of Platt Scaling – is surprisingly effective at calibrating predictions."

Relevance to MedOracle: this is the source for the entropy-based confidence score $c = 1 - H(P)/\log K$ used to gate each modality's contribution before fusion, and for treating a peaked probability distribution as more trustworthy than a flat one.